

DU Mathsoc Problem Solving: Set 4

Date: 06/10/2026

~Beginners Paper~

For any assistance or inquiries about the problem sets, please email me at: quizmaster.maths@csc.tcd.ie.

Problem 1. An island has 15 green, 7 white, and 11 black chameleons. When two chameleons of different colours meet, they both change to the third colour. Can all chameleons ever become the same colour?

Problem 2. Let ABC be a triangle inscribed in circle ω . Show that $AC \perp BC$ if and only if AB is a diameter of ω .

Problem 3. Let O and H denote the circumcentre and orthocentre of an acute triangle ABC , respectively. Show that $\angle BAH \equiv \angle CAO$.

Problem 4. For $n \geq 0$ natural, find $I_n = \int_0^{\frac{\pi}{2}} (\cos x)^n \cdot \cos(nx) dx$.

Problem 5. Let $\alpha_1, \alpha_2, \dots, \alpha_n$ be distinct real numbers. Show that $e^{\alpha_1 t}, e^{\alpha_2 t}, \dots, e^{\alpha_n t}$ are linearly independent as functions over $\mathbb{R}$.